Kai **Zhu**

PH. D. STUDENT · ZHEJIANG UNIVERSITY

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Summary

I am a Ph.D. student in Pharmaceutics at Zhejiang University, advised by Prof. Tingjun Hou. I am also a visiting student at the Italian Institute of Technology, working with Luigi Bonati and Prof. Michele Parrinello. My research focuses on developing machine learning–based enhanced sampling methods and their applications to atomic systems.

Education

College of Pharmaceutical Sciences, Zhejiang University — *Hangzhou, China*

PH.D. IN MEDICINE CHEMISTRY — *Sep. 2023 – Jun. 2028*

College of Pharmaceutical Sciences, Wenzhou Medical University — *Wenzhou, China*

B.S. IN PHARMACEUTICAL SCIENCE — *Sep. 2019 – Jun. 2023*

Research Experience

Italian Institute of Technology — *Genova, Italy*

RESEARCH INTERN, ADVISED BY **PHD. LUIGI BONATI** AND **PROF. MICHELE PARRINELLO** — *Dec. 2025 – Dec. 2027*

Research Topics: machine learning, enhanced sampling, atomic simulation

Zhejiang University — *Hangzhou, China*

PHD STUDENT, ADVISED BY **PROF. TINGJUN HOU** AND **PROF. DAN LI** — *Jun. 2023 – Jun. 2025*

Research Topics: machine learning, enhanced sampling, intrinsically disordered proteins

Honors & Awards

2026 **Young Science and Technology Talent Cultivation Program (Doctoral Student Special Plan)**, China Association for Science and Technology (CAST) — *China*

2025 **Qiu Shi Flying Eagle Scholarship**, Zhejiang University — *Hangzhou, China*

Selected Publications

† Co-first author; * Corresponding author;

2026 **Contrastive Learning of Dynamical Representations for Enhanced Molecular Sampling** [4]
K. Zhu†, J. Zhang†, P. Novelli, T. Hou*, L. Bonati*. **arXiv** arXiv:2606.15495 (2026).

2026 **Targeting the Intrinsically Disordered AR-NTD through a Machine Learning-Based Enhanced Sampling Workflow** [3]
K. Zhu†, H. Wang†, J. Zhang†, R. Hu, L. Jiang, H. Zhang, Y. Kang, T. Hou*, D. Li*. **Nat. Commun.** (2026).

2026 **A Scalable and Quantum-Accurate Foundation Model for Biomolecular Force Field via Linearly Tensorized Quadrangle Attention** [2]
Q. Su†, **K. Zhu†**, Q. Gou†, ..., J. Wang*, C. Hsieh*, T. Hou*. **Nature Communications** 17, 3639 (2026).

2025 **Enhanced Sampling in the Age of Machine Learning: Algorithms and Applications** [1]
K. Zhu†, E. Trizio†, J. Zhang, R. Hu, L. Jiang, T. Hou*, L. Bonati*. **Chemical Reviews** 126 (1), 671-713.

Skills

Programming Python, C++, C, LaTeX, Linux
Frameworks PyTorch, TensorFlow, NumPy, Git, Anaconda
Atomic simulation GROMACS, PLUMED, AMBER, OpenMM, ASE, LAMMPS, CP2K